

CHEBOKSARY, Russian Federation

WMO: 275810

Lat: 56.083N

Lon: 47.333E

Elev: 163

StdP: 99.38

Time Zone: 3.00 (E03)

Period: 08-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-25.2	-22.9	-28.8	0.3	-25.3	-25.9	0.4	-22.8	10.7	-5.7	10.0	-5.8	2.7	80	0.622

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	9.7	31.1	18.9	29.0	18.7	27.1	18.2	20.9	26.5	19.9	25.8	19.2	25.4	3.2	210

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
19.0	14.1	23.2	18.0	13.2	22.3	17.1	12.4	21.5	60.9	26.3	57.9	25.8	55.3	25.1	23.8

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 9.8	8.8	8.0	-27.2	32.4	2.7	2.7	-29.2	34.4	-30.7	35.9	-32.2	37.4	-34.2	39.3		
(5)			-27.9	22.3	2.9	1.4	-30.0	23.3	-31.7	24.1	-33.3	24.9	-35.4	25.9		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	5.0	-11.7	-9.5	-4.1	5.5	14.4	17.6	20.2	18.4	12.1	4.7	-1.4	-6.8
	DBStd	12.11	6.66	6.39	4.67	4.56	4.71	4.25	3.70	4.43	4.13	4.54	4.57	6.31
	HDD10.0	2884	673	546	438	150	14	1	0	1	25	174	342	521
	HDD18.3	5054	931	779	697	386	138	63	21	53	191	423	592	779
	CDD10.0	1068	0	0	0	14	151	230	316	261	87	8	0	0
	CDD18.3	197	0	0	0	0	17	43	78	56	3	0	0	0
	CDH23.3	1722	0	0	0	1	155	381	687	477	22	0	0	0
	CDH26.7	533	0	0	0	0	24	112	227	168	1	0	0	0
Wind	WSAvg	4.1	4.4	4.6	4.7	4.3	4.0	3.7	3.2	3.2	3.5	4.4	4.6	4.6
Precipitation	PrecAvg	566	38	29	33	30	38	67	71	61	56	56	45	38
	PrecMax	664	77	57	72	70	71	160	146	117	115	109	84	60
	PrecMin	401	14	9	5	6	11	19	9	2	15	11	7	10
	PrecStd	66	14	13	16	18	16	37	33	28	27	26	18	12
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	1.2	2.1	7.0	22.1	29.1	32.1	35.2	35.1	26.0	18.1	10.2	5.2
		MCWB	0.9	0.9	3.4	12.8	16.9	17.5	18.7	19.2	16.4	12.5	8.9	4.9
	2%	DB	0.8	1.1	4.2	19.0	27.0	29.8	31.8	31.0	23.2	14.9	8.2	3.1
		MCWB	0.6	0.6	1.6	10.6	16.3	19.6	18.8	18.6	15.3	11.1	7.3	2.6
	5%	DB	-0.2	0.1	3.0	15.9	25.0	27.8	29.0	28.0	20.9	12.9	6.8	1.8
		MCWB	-0.5	-0.2	1.2	9.2	15.3	18.5	18.7	18.5	14.6	9.8	6.0	1.4
	10%	DB	-2.1	-0.9	1.9	13.2	22.9	25.2	27.0	25.9	18.8	11.0	4.2	0.9
		MCWB	-2.5	-1.4	0.7	7.4	14.7	17.3	18.5	17.9	13.5	8.8	3.6	0.5
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	0.7	1.2	3.7	13.1	18.8	22.2	21.8	21.7	17.8	13.2	9.3	5.1
		MCDB	0.8	1.3	6.1	20.0	25.9	28.3	27.3	26.2	23.1	16.2	10.0	5.1
	2%	WB	0.2	0.4	2.1	11.7	17.6	20.9	20.6	20.4	16.4	11.7	7.3	2.5
		MCDB	0.3	0.7	3.2	17.3	24.8	26.6	26.3	26.4	21.2	14.3	8.2	2.7
	5%	WB	-0.9	-0.2	1.4	9.9	16.4	19.5	19.9	19.6	15.3	10.4	5.6	1.1
		MCDB	-0.6	0.1	2.4	14.9	22.6	25.1	25.6	26.0	19.5	12.2	6.3	1.3
	10%	WB	-2.8	-1.5	0.7	8.1	15.2	18.3	19.2	18.7	14.2	9.1	3.7	0.2
		MCDB	-2.5	-1.1	1.7	12.1	21.5	23.9	25.4	24.6	17.6	10.7	4.1	0.4
Mean Daily Temperature Range	5% DB	MDBR	5.5	5.7	6.2	8.1	10.4	9.5	9.7	9.3	7.6	5.4	3.9	4.5
		MCDBR	4.3	4.4	6.3	12.5	12.6	11.9	12.3	12.1	11.1	8.2	5.1	3.5
		MCWBR	4.4	4.4	4.7	6.7	5.9	5.3	4.9	4.6	5.3	5.3	4.8	3.5
	5% WB	MCDBR	4.5	4.1	5.5	11.8	11.4	10.8	10.0	10.4	9.6	7.1	4.9	3.3
		MCWBR	4.6	4.1	4.6	6.6	6.0	5.4	4.9	4.7	5.1	5.4	4.8	3.5
Clear-Sky Solar Irradiance	taub	0.285	0.316	0.355	0.402	0.380	0.375	0.399	0.423	0.369	0.347	0.309	0.281	
	taud	2.081	2.067	2.034	2.143	2.346	2.399	2.338	2.245	2.383	2.339	2.215	2.062	
	Ebn at Noon	586	719	785	809	860	869	836	783	782	695	564	481	
	Edh at Noon	62	94	125	131	114	110	114	118	90	73	55	49	
All-Sky Solar Radiation	RadAvg	0.61	1.50	3.00	4.09	5.46	5.81	5.73	4.47	2.72	1.25	0.49	0.33	
	RadStd	0.09	0.26	0.32	0.41	0.37	0.48	0.52	0.43	0.34	0.20	0.09	0.08	

Historical Trends

		DBAvg	Heating			Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP		HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	
(47) Regional (6 neighbors)		+0.53	N/A	N/A	N/A	N/A	N/A	-142	-192	N/A	N/A	

Nomenclature: See separate page